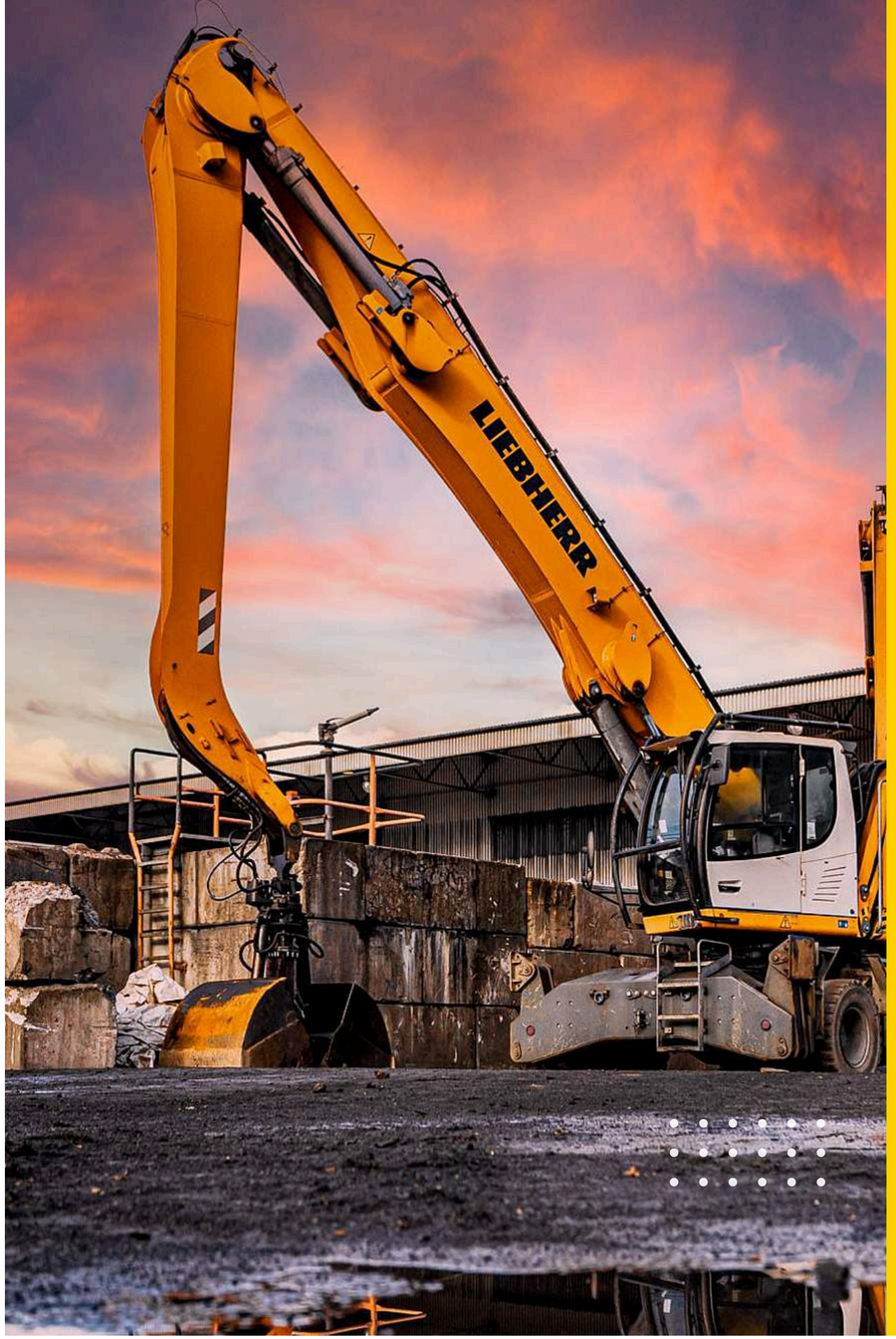


HAT1 FINANCIAL REPORTING · 2026

Liebherr Group

Financial Reporting and Investment Analysis 2022–2025

CENTRAL QUESTION

How financially strong is Liebherr, and which factors will drive its future performance?

For a long-term investor: take a minority stake now – or not yet?

Benchmark: Caterpillar Inc. and the average of 105 listed US machinery companies

PRESENTED BY

Simon Beermann · Adam Fliegel · Zofia Narkiewicz · Philipp Hoffmann · Zarah Gilmore Islam

Fontys University of Applied Sciences, Venlo · International Finance and Control · Lecturer: Kim Verspeek

THE ANSWER FIRST

Financially sound – but not yet an adequate return

Four findings decide the investment case

STRENGTH

53.5 %

Equity ratio 2025

Cash and current financial assets cover current debt **2.6 times**.

WEAKNESS

2.8 %

Operating margin 2025

Less than a fifth of Caterpillar (17.2 %) and the industry (15.9 %).

WEAKNESS

–€176 m

Free cash flow 2025

Negative in three of four years – investment runs ahead of cash generation.

ACCOUNTING

1.6–2.8 %

Margin range in our tests

The measurable accounting effects point to a **weaker**, not a stronger result.

Not yet. **Do not invest at present – reassess the case on the 2026 financial statements.**

Source: Report, Executive summary and chapters 7–8; Excel workbook, sheet Dashboard

COMPANY AND ENVIRONMENT

Diversification kept Liebherr growing while construction weakened

€14,772 m
Revenue 2025, +1.0 %

55,963
Employees

13
Product segments

1949
Family-owned · Bulle (CH)



Revenue 2025 by sector

63.3 % · Construction & mining · −5.5 %

36.7 % · Other · +14.7 %

€9,345 m vs. €5,427 m – components alone grew 35.5 %

External pressures

- **Demand:** European construction output forecast to grow only about 2 % a year until 2028
- **Prices and costs:** US tariffs and low-cost competitors – reported by Liebherr for its cranes
- **Financing:** ECB main refinancing rate 2.65 % (Sept. 2026) – machines cost customers more to finance

LIEBHERR'S OWN OUTLOOK

Orders at the start of 2026 above the previous year – further growth expected, but **no significant upswing before 2027**. US tariff policy remains a risk.

Source: Report, chapters 2 and 3 (§3.1.1) and §5.7; Liebherr-International AG (2026); EUROCONSTRUCT (2026); European Central Bank (2026)

How Liebherr creates competitive advantage

Differentiation in custom segments, scale in volume segments – financed by patient family capital



Diversified product portfolio

13 segments, from mining excavators to aerospace components, spread the market risk.

+14.7 %

other segments 2025, while construction & mining fell 5.5 %

THE PRICE

Stabilises revenue, but does not raise the margin.



Vertical integration

Key technologies built in-house; own sales and service companies instead of dealers.

29.1 %

PP&E in % of revenue (Caterpillar 22.4 %)

THE PRICE

Capital tied up in assets and stock – inventories 42.9 % of revenue.



Innovation & research

Research and development rose every year, fully expensed, financed from retained profit.

€708 m

R&D 2025 = 4.8 % of revenue (Caterpillar 3.2 %)

THE PRICE

Spending kept rising while operating profit fell 39 % in 2025.

WHAT IT MEANS

The advantage is real – but it sits in the asset base, not yet in the margin.

For the investor the question is not *whether* the advantage exists, but whether it is turned into returns.

Only €56 m of a €272 m profit was distributed in 2025.

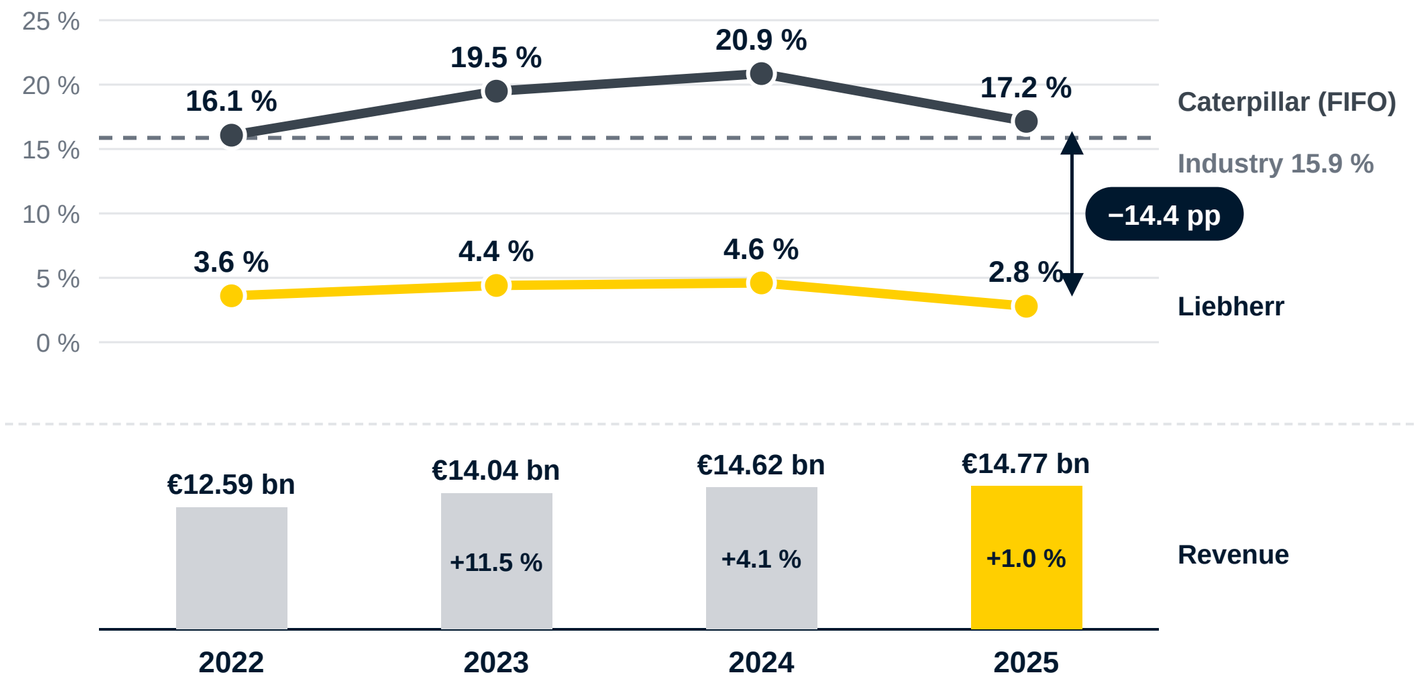
Source: Report §3.2–3.3, Appendix E (VRIO); Liebherr-International AG (2023–2026); Caterpillar Inc. (2026)

FINANCIAL PERFORMANCE

Growth has stalled and the margin fell to 2.8 %

Revenue still grows, but barely – and less of it turns into profit

Operating margin (%) and revenue (€ bn), 2022–2025

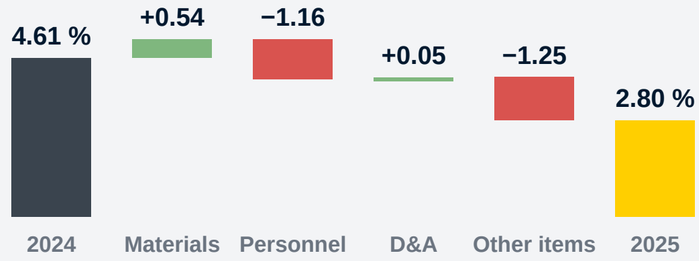


DuPont: why the return on equity fell



pre-tax return on equity · contributions in pp

Margin bridge 2024 → 2025



contribution in percentage points of revenue

INVESTOR LENS **Each percentage point of margin is worth about €148 m of operating profit**

Source: Report §5.2–5.3, Figures 1, 2a and J2; Excel workbook, sheets D4, D6, D10 and Dashboard. Caterpillar on the FIFO basis; industry = Damodaran (2026c), 2025 reference level

ACCOUNTING ANALYSIS

Key figures rest on judgements we cannot verify

Most room for judgement: inventories, provisions and fixed assets – and one cost line without any note

REPORTING LIMIT **Summary statements only (ISA 810) · full audited statements not public · no segment note (IFRS 8 not applicable)**

INVENTORIES · IAS 2

€6,341 m

36.3 % of total assets

- Overhead allocated at a “normal capacity” that is not defined
- Allowance for obsolete stock not disclosed

PROVISIONS · IAS 37

€864 m

of which warranty €530 m

- Gross reversals rose from €90 m (2022) to €143 m (2025)
- But the net charge stayed flat: €195 m → €194 m

PP&E · IAS 16

€4,304 m

24.6 % of total assets

- Depreciation 5.5 % of cost vs. 6.6 % at Caterpillar
- Own work capitalised €317 m = 77 % of the operating result

DISCLOSURE GAP

+€228 m

other operating expenses 2025 (+11.7 %)

- The largest cost increase of 2025 has no note at all
- 4 of 9 red-flag tests raised a flag

INVESTOR LENS **The 2025 operating result is not a reliable base for extrapolation**

Source: Report §4.1–4.4, Appendix G (Table G2) and Appendix H (Table H1); Excel workbook, sheets C2 and D9; IFRS Foundation (n.d.)

Adjusting for comparability widens the gap

Actual adjustment (Caterpillar) vs. illustrative sensitivities (Liebherr)

ACTUAL ADJUSTMENT · BOOKED IN THE WORKBOOK

Caterpillar: inventories restated LIFO → FIFO

IAS 2 prohibits LIFO and Liebherr does not use it. The disclosed LIFO reserve makes the correction exact, not estimated.

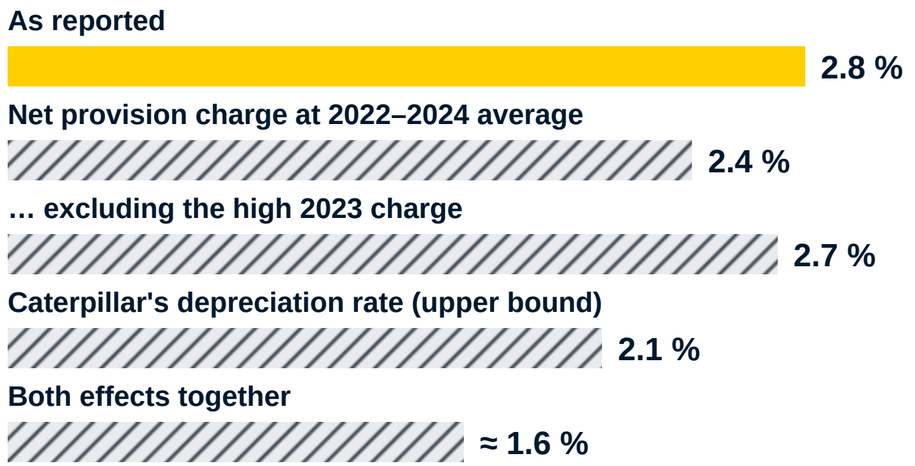
Inventories (LIFO)	+	LIFO reserve	=	FIFO basis
$18,135 m		$4,305 m		$22,440 m

Equity (before tax)	Operating profit 2025	Operating margin
+$4,305 m	+$441 m	16.5 → 17.2 %

Also reclassified: Caterpillar's 2022 goodwill impairment of $925 m as non-recurring. At equal depreciation the gap would be up to **15.1 pp** instead of 14.4 pp.

ILLUSTRATIVE SENSITIVITY · NOT BOOKED

Liebherr operating margin 2025 under the findings



Own work capitalised cannot be measured. The equity ratio would still be **50.4 %** at Caterpillar's depreciation rate – the balance sheet strength holds.

INVESTOR LENS **Accounting does not rescue the case – the reported 2.8 % is, if anything, flattered**

Source: Report §4.4, §6.1, Table J9 and Table H3; Caterpillar Inc. (2023, 2026); Excel workbook, sheets C1 (block A), D9 and D11. Caterpillar figures in USD m, Liebherr in € m – never combined

A strong balance sheet is financing a cash shortfall

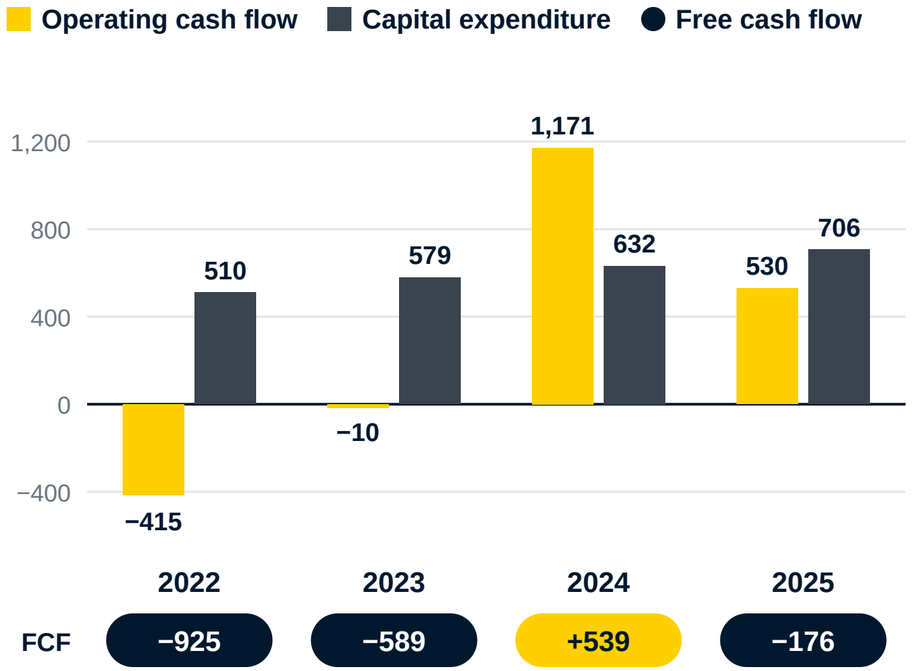
Liquidity and solvency are solid – but investment runs ahead of operating cash flow

53.5 %
Equity ratio 2025
Bank and lease liabilities 23.7 % of capital vs. 39.3 % industry

2.6×
Cash cover of current debt
€3,147 m cash and current financial assets vs. €1,215 m

2.90×
Debt / EBITDA 2025
Up from 2.14 in 2024 – above the industry's 2.30

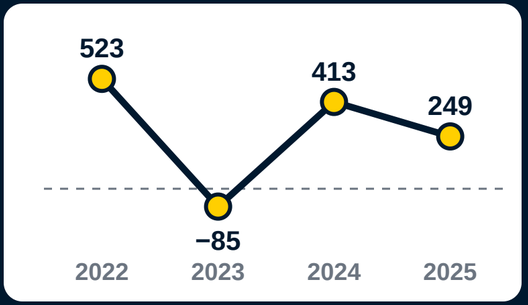
Operating cash flow vs. capital expenditure, € m



2022–2025 CUMULATIVE

Capex **€2,427 m** vs. operating cash flow **€1,276 m** = **190 %**

NET FINANCIAL POSITION, € m



Covers the 2025 shortfall after distributions (€232 m) for only **≈ 1.1 years**.

INVESTOR LENS **Low risk of distress – but the cushion is shrinking**

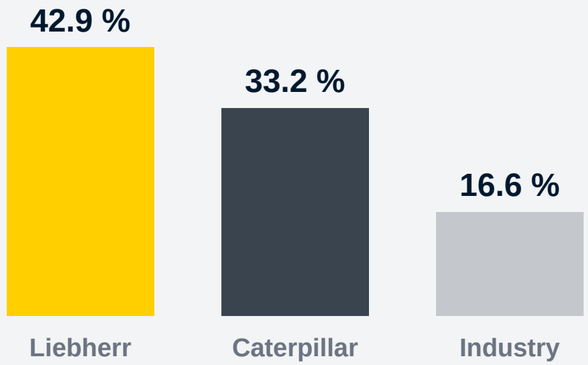
Source: Report §5.5–5.6, Figures 3 and 4, Table 2; Excel workbook, sheets D3, D5, D7 and Dashboard; industry: Damodaran (2026b)

WORKING CAPITAL

Inventories tie up 42.9 % of revenue

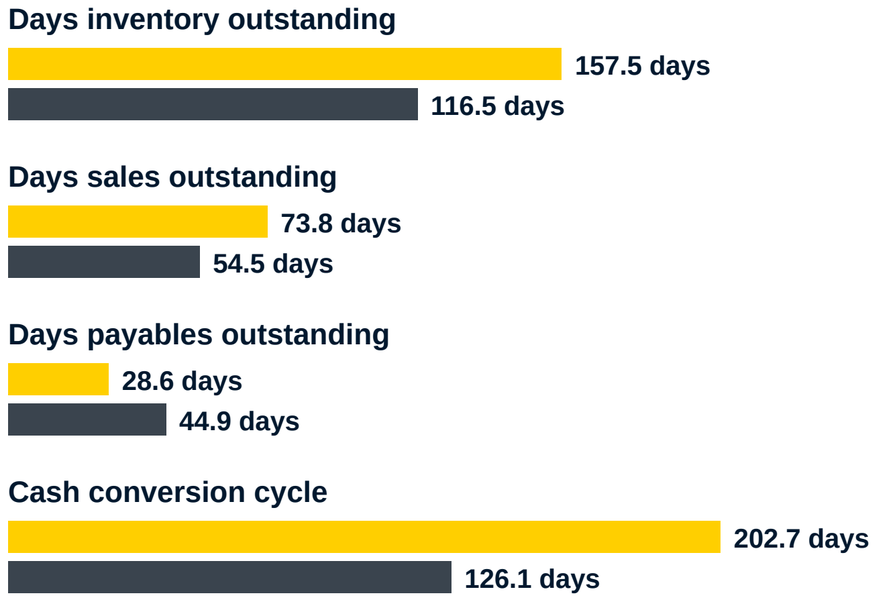
2.6 times the industry average – the cash conversion cycle is a revenue-based proxy

Inventories in % of revenue, 2025



- Causes: in-house production, long order cycles, stock held by own sales companies
- Finished goods alone: €2,490 m

Working-capital days 2025 (revenue basis)



THE LEVER

€405 m

released by a cycle **10 days shorter** (€40.5 m revenue a day)

… but only once. A lasting turnaround needs margin.

INVESTOR LENS **77 days longer cash cycle than Caterpillar – limits Liebherr’s financial flexibility**

Source: Report §5.4, Figure 2b, Table J5; Excel workbook, sheets D3 (block E) and Dashboard; industry: Damodaran (2026e). Days on a revenue basis – a proxy, as Liebherr reports no cost of sales

What would have to change: three conditions for a reassessment

Check them on the 2026 statements – a new assessment only if all three are met. In 2025 none of them is.



1 **Operating margin of at least 3.9 % (four-year average)**

Status 2025 — × 2.8 %

2 **Free cash flow positive and, over time, covering distributions**

Status 2025 — × −€176 m

3 **Information rights: audited statements and other operating expenses**

Status 2025 — × Not secured

REWARD

Resilient, diversified group · back to the 3.9 % average margin = **+€157 m** operating profit at 2025 revenue

RISK AND LIMITS

ROCE 3.5 % (pre-tax) vs. 7.70 % benchmark – needs a margin of ~6–8 % · minority stake: little influence, low payout, hard to sell

Even then: meeting the conditions triggers a new analysis, not a purchase – a minority stake in an unlisted group would only make sense at a discounted price.

Source: Report §5.3, §5.7 and chapter 8, Table 3; Excel workbook, sheets D3, D10 and D11; benchmark: Damodaran (2026a)



CONCLUSION

Financially strong. Operationally challenged.

1 **The balance sheet is strong – and stays strong**

53.5 % equity ratio and 2.6× cash cover; still ≈ 50 % equity ratio in our depreciation sensitivity.

2 **Returns and cash generation are too weak**

2.8 % margin, 12.5–16.2 pp below Caterpillar in every year; free cash flow negative in 3 of 4 years.

3 **The accounting sharpens the verdict**

The measured effects point to a margin of 1.6–2.8 %, and key estimates cannot be verified from outside.

Not yet: the investor should see the turn in the figures before committing capital.



Thank you.

Simon Beermann · Adam Fliegel · Zofia Narkiewicz · Philipp Hoffmann · Zarah Gilmore Islam · HAT1 Financial Reporting · Fontys Venlo

Images: Liebherr-International AG

BACKUP

Backup slides for the questions

Not part of the 10-minute talk – click a slide to jump there; ◂ INDEX on every slide brings you back

A Ratios, workbook and what-if — MOST ASKED

14 Profitability ratios ▸
15 Efficiency, leverage, liquidity ▸
16 Workbook to ratio ▸
17 What-if I ▸
18 What-if II ▸
19 Provisions ▸
20 Sensitivity ▸
21 Long list to short list ▸
22 Key accounts ▸
23 Red-flag screen ▸
24 Short list I ▸
25 Short list II ▸
26 Workbook ▸

B Business model and environment

27 DESTEP ▸
28 Five forces ▸
29 VRIO and SWOT ▸

C Accounting analysis

30 Red flags ▸
31 Room for judgement ▸
32 Adjustments and limits ▸
33 Reliability of ratios ▸

D Financial analysis

34 Growth and costs ▸
35 DuPont ▸
36 Liquidity and benchmark ▸
37 Cash flow ▸

E Recommendation, scenarios and risks

38 Why 3.9 % ▸
39 Scenarios ▸
40 Risks and drop trigger ▸

F Likely questions and answers

41 Questions 1/2 ▸
42 Questions 2/2 ▸
43 Questions 3/3 ▸

44 References (APA) ▸

◂ INDEX

Profitability: returns are low because the margin is low

Margins and returns on capital – formula, Liebherr and Caterpillar, 2025

GROUP	RATIO	FORMULA	LIEBHERR	CATERPILLAR
PROFITABILITY margins	Operating margin	EBIT / revenue	2.8 %	17.2 %
	Pre-tax margin	Profit before tax / revenue	3.0 %	17.9 %
	Net profit margin	Profit after tax / revenue	1.8 %	13.8 %
	EBITDA margin	EBITDA / revenue	6.8 %	19.8 %
PROFITABILITY returns on capital	ROA	EBIT / average total assets	2.4 %	11.9 %
	ROCE	EBIT / (avg. equity + avg. non-current debt)	3.5 %	20.2 %
	ROE	Profit after tax / average equity	3.0 %	38.1 %*
	Pre-tax ROE	Profit before tax / average equity	4.8 %	49.4 %

* Not comparable – treasury shares reduce Caterpillar's equity. ROA is measured on EBIT, because Liebherr's interest expense is not disclosed.

DUPONT

ROE = net margin × asset turnover × leverage

leverage = average total assets / average equity

= **profitability** × **efficiency** × **leverage** – the ratio groups of slides 14 and 15

	MARGIN	TURNOVER	LEVERAGE	ROE
2025	1.84 %	× 0.85	× 1.89	**3.0 %**
pre-tax	2.96 %	× 0.85	× 1.89	**4.8 %**
2023 pre-tax	4.76 %	× 0.88	× 1.81	**7.6 %**

Leverage works on ROE, **not on the EBIT margin** – EBIT is measured before interest.

Source: Excel workbook, sheets D3 (blocks C and D) and D4; Report §5.3, Table J14. 2025; Caterpillar on the FIFO basis

◂ INDEX

Efficiency, leverage, liquidity: slow capital, solid finance

How capital is used, how it is financed and how liquid Liebherr is – 2025

GROUP	RATIO	FORMULA	LIEBHERR	CATERPILLAR
EFFICIENCY asset utilisation · working capital	Asset turnover	Revenue / average total assets	0.85	0.69
	Fixed asset turnover	Revenue / average PP&E	3.49	4.74
	Days inventory outstanding	Avg. inventories / revenue × 365	157.5	116.5
	Days sales outstanding	Avg. receivables / revenue × 365	73.8	54.5
	Days payables outstanding	Avg. payables / revenue × 365	28.6	44.9
	Cash conversion cycle	DIO + DSO - DPO	202.7	126.1
LEVERAGE solvency	Equity ratio	Equity / total assets	53.5 %	24.9 %
	Leverage (DuPont)	Avg. total assets / avg. equity	1.89	3.97
	Debt / equity	Debt incl. pensions / equity	0.40	1.84
	Debt / capital	Bank + lease liab. / (those + equity)	23.7 %	64.8 %
	Debt / EBITDA	Bank + lease liabilities / EBITDA	2.90	3.52
LIQUIDITY short term	Current ratio	Current assets / current liabilities	2.31	1.55
	Quick ratio	(Current assets - inventories) / current liab.	1.16	0.94
	Cash cover	Cash + current fin. assets / current debt	2.6×	–

READ WITH CARE

Days = efficiency, but they drive liquidity: 203 days of revenue must be financed.

Revenue basis: Liebherr reports no cost of sales; on its own cost basis DIO would be 299 days – not comparable.

Two debt measures: “debt” includes pensions (€3,714 m); bank and lease liabilities (€2,898 m) match the Damodaran benchmark.

Interest cover: not computable – no interest expense disclosed. Indicator: operating profit / interest paid 3.90.

Source: Excel workbook, sheets D3 (blocks D, E, G, H), D4 and D7; Report Tables J5 and J14. 2025; days on average balances; Caterpillar on the FIFO basis

◂ INDEX

From workbook to ratio: every figure is a formula

Excerpt of sheet D3 – inputs link to the recast statements, ratios link to the inputs

Liebherr-Caterpillar-Restructured-financial-statements.xlsx · sheet D3 Ratios

B40 | *fx* `=B10/B27`

	A	B · 2025	formula in column B
7	Revenue	14,772	`='B1 Income statement by nature'!B5`
8	Operating profit (EBIT)	413	`='C1 Accounting adjustments'!B30`
10	Profit after tax	272	`='B1 Income statement by nature'!B18+'C1…'!B27`
14	Total equity	9,354	`='C1 Accounting adjustments'!B21`
16	Inventories	6,341	`='C1 Accounting adjustments'!B13`
19	Current assets	12,709	`='B2 Restructured balance sheet'!B17+'C1…'!B12`
20	Current liabilities	5,498	`='B2 Restructured balance sheet'!B33`
27	Average total equity	9,128	`=(B14+C14)/2`
29	Average inventories	6,373	`=(B16+C16)/2`
35	Operating margin	2.80 %	`=B8/B7`
40	Return on equity (ROE)	**2.98 %**	`=B10/B27`
49	Days inventory outstanding	157.5	`=B29/B7*365`
60	Current ratio	2.31	`=B19/B20`

Return on equity `=B10/B27`

272 / 9,128 → **3.0 %**

Days inventory outstanding `=B29/B7*365`

6,373 / 14,772 × 365 → **157.5 days**

Current ratio `=B19/B20`

12,709 / 5,498 → **2.31**

0 open differences on sheet E2

Inputs come from the recast statements B1/B2 and the adjustments on C1 – nothing is typed in.

Source: Excel workbook, sheet D3 Ratios (cell formulas as in the file), column B = Liebherr 2025, € m; checks on sheet E2

◂ INDEX

What-if I: direction matters more than the formula

Balance sheet and liquidity – what happens to the ratios

Suppliers paid in cash	Current ratio ▲ Cash ▼	Both sides fall by the same amount, so a ratio above 1 rises: €500 m paid → 2.31 to 2.44 (illustration).
Inventory write-down	EBIT ▼ Current ratio ▼ DIO ▼	The days **look** better while the position is weaker – the allowance is not disclosed.
New bank debt for capex	Equity ratio ▼ Debt/EBITDA ▲ Leverage ▲	EBIT margin unchanged; net margin falls through interest. ROE rises only if the return beats the interest rate.
Cash conversion cycle −10 days	Cash ▲ Margin –	Releases about €405 m once (€40.5 m revenue a day) – not a recurring improvement.
Receivables collected faster	DSO ▼ OCF ▲ Current ratio –	Cash replaces receivables inside current assets, so liquidity ratios barely move.

INVESTOR LENS **A better ratio is not always a better company**

Source: Report §5.4–5.5; Excel workbook, sheets D3 and D7. Directions are logical consequences; the €500 m case is an illustration on 2025 balances

◂ INDEX

What-if II: accounting choices move profit, not cash

Effect of an accounting choice on EBIT, equity and cash

CHANGE	EBIT	EQUITY	CASH / OCF	LIEBHERR EXAMPLE
Provision addition	▼	▼	–	Expense now – the obligation is estimated
Provision usage (utilisation)	–	–	▼	Obligation settled – no profit effect
Provision reversal (release)	▲	▲	–	Estimate no longer needed – profit without cash
Slower depreciation (longer life)	▲	▲	–	Liebherr 5.5 % vs. Caterpillar 6.6 % of cost
More own work capitalised	▲ **now**	▲	–	€317 m = 77 % of 2025 EBIT; returns as depreciation
LIFO → FIFO, rising prices (CAT)	▲	▲	–	+$441 m EBIT, +$4,305 m equity before tax

INVESTOR LENS **Operating cash flow is the check on the accruals – only 28.4 % of EBITDA in 2022–2025**

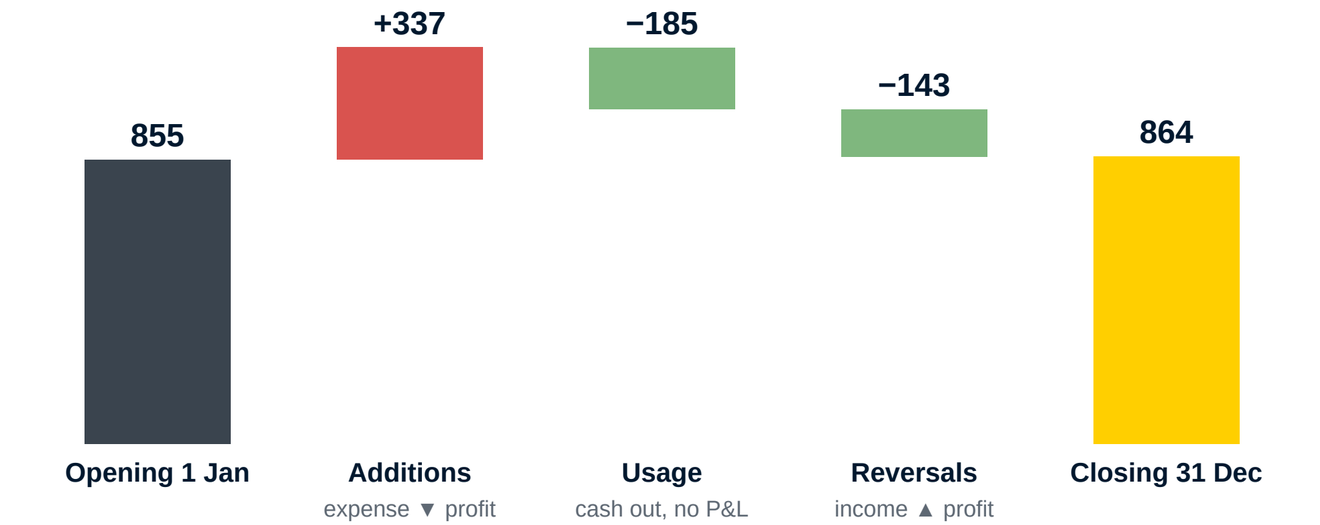
Source: Report §4.2–4.4 and §6.2; Excel workbook, sheets A2 (note 10), C1, C2 and D9

◂ INDEX

Provisions: reversals rose – but so did additions

Movement of provisions 2025 (IAS 37), € m – of which warranty 530

Roll-forward 2025, all provisions



Net charge = additions − reversals	2022	2023	2024	2025
	193	338	195	194

How to read it

- **IAS 37:** present obligation from a past event, probable outflow, reliable estimate
- **Usage ≠ reversal:** only additions and reversals touch profit
- **Hidden reserve test:** a release to support profit would show as a *falling* net charge – 195 → 194, so not visible
- **Flag kept:** other provisions built up by €146 m in 2023 are being released (€63 m in 2025); the notes cannot show why

Source: Excel workbook, sheets A2 (note 10, rows 68–80), C2 and D11 (block B); Report §4.2 and §4.4, Table H1 test 1. € m

◂ INDEX

Sensitivity: every measured effect lowers the margin

Liebherr operating margin 2025 – illustrations on published figures, not booked adjustments

STEP	CALCULATION, € M	EBIT	MARGIN
As reported	–	413	2.80 %
Net provision charge at the 2022–2024 average	413 - (1.76 % - 1.31 %) × 14,772	347	2.35 %
… without the high charge of 2023	413 - (1.43 % - 1.31 %) × 14,772	395	2.68 %
Caterpillar's depreciation rate (6.6 % instead of 5.5 % of cost)	413 - 105	308	2.08 %
Both effects together	413 - 66 - 105	242	1.64 %

2.80 − 0.44 − 0.71 = 1.64 %

Sheet D9 also shows a **gross-reversal** view (2.5 % at average reversals, 1.8 % without any reversal). The report uses the **net charge** (D11), because additions rose together with the reversals. Own work capitalised (€317 m) cannot be measured.

Source: Excel workbook, sheets D9 (blocks A and B) and D11 (block B); Report §6.1, Tables H3 and J13

◂ INDEX

From 30 treatments to 22 findings

How the long list became the short list – four steps

STEP 1 · TABLE G2

Long list

30

accounting treatments with room for choice

▶

STEP 2 · TABLE G1

Key treatments

9

key account × high or medium-to-high flexibility

▶

STEP 3 · TABLE H1

Red-flag screen

9

8 tests + 1 consistency check → 4 flags · 3 clean · 2 not testable

▶

STEP 4 · TABLE H2

Short list

22

findings → 7 not adjustable · 5 reversed · 10 not reversible

The nine key treatments

Inventory cost and write-down · own work capitalised · impairment · expected credit losses · three provision categories · deferred taxes. Revenue timing and the by-nature format are added – they affect every ratio.

Left out – and why

Borrowing costs, lessee accounting and currency translation: the standard prescribes the mechanics. Development costs: never capitalised, kept only for the comparison. The two “9”s are different counts.

Source: Report §4.2–4.4, Tables G1, G2, H1 and H2

◂ INDEX

Seven key accounts carry the judgement

Key accounts × flexibility = the nine key treatments of the long list

KEY ACCOUNT	2025, € M	STANDARD AND JUDGEMENT	FLEXIBILITY	WHY IT IS ON THE SHORT LIST
Inventories long list B1, B2	**6,341** 42.9 % of revenue	**IAS 2** – Fixed overheads absorbed at an undefined normal capacity; allowance not disclosed	High	Largest account – 36.3 % of total assets; quality cannot be verified
Provisions long list C1–C3	**864** of which warranty 530	**IAS 37** – Recognition, measurement and reversal of the estimate	High	Reversals €143 m = 34.6 % of the operating result if credited within it
PP&E and rental fleet long list B5–B7	**4,304** 24.6 % of total assets	**IAS 16 / IAS 36** – Useful lives, residual values, indicator-based impairment	High (impairment), medium (lives)	Depreciation 5.5 % of cost vs. 6.6 % at Caterpillar
Own work capitalised long list B3	**317** 77 % of the operating result	**IAS 16 / IAS 38** – Which production cost is capitalised rather than expensed	Medium to high	Moves cost out of the year – returns later as depreciation
Trade receivables and contract assets long list B11	**3,116** 17.8 % of total assets	**IFRS 9** – Simplified lifetime expected credit loss through a provision matrix	Medium	Grew 9.1 % against revenue +1.0 % in 2025; allowance not disclosed
Revenue long list A1, A2	**14,772**	**IFRS 15** – Point in time against over time; cost-to-cost estimate	Medium	Share recognised over time not disclosed – affects every ratio
Deferred tax assets and tax charge long list D1	**285** tax rate 37.8–99.2 %	**IAS 12** – Probability of future taxable profit; announced instead of enacted rates	High	No tax reconciliation – after-tax ratios not usable

Source: Report §4.2–4.3, Tables G1 and G2; Liebherr-International AG (2026); IFRS Foundation (n.d.)

◂ INDEX

Red-flag screen: nine tests, four flags

4 raised · 3 clean · 2 not testable – reasons to ask, not proof of manipulation

#	TEST	EVIDENCE 2022–2025	VERDICT
1	**Unexplained transactions that boost profits**	Provision reversals 90 → 143 (2022–2025) = 34.6 % of 2025 EBIT; net charge flat (195 → 194); credited line not disclosed	Raised
2	**Inventories or receivables rise faster than sales**	Inventories +27.3 % vs. revenue +8.2 % (2022); receivables +9.1 % vs. revenue +1.0 % (2025)	Raised twice
3	**Gap between profit and cash flow or taxable profit**	Operating cash flow −415 and −10 (2022–23) against profit; tax rate 99.2 % → 37.8 %, no reconciliation	Raised
4	**R&D partnerships, special purpose entities, sold receivables**	None disclosed in four years; R&D fully expensed	No flag
5	**Unexpected large asset write-offs**	PP&E impairment 8, 55, 9, 3 – 2023 six times the other years, same one-sentence explanation	Raised for 2023
6	**Large year-end adjustments**	No interim figures published, so no within-year comparison	Not testable
7	**Qualified opinions or auditor changes**	Ernst & Young all four years, no qualification, no restatement (ISA 810 opinion)	No flag
8	**Related party transactions**	No note in four years – family-owned group that also runs six hotels	Not testable
9	**Changes in accounting policies or estimates**	Policies worded identically; new standards without material effect	No flag

Source: Report §4.4, Table H1 (8 tests of step 5 + consistency check of step 3); Liebherr-International AG (2023–2026)

◂ INDEX

Short list I: what cannot be fixed – and what we fixed

Findings 1–12 of 22 (Table H2)

#	FINDING	ACCOUNT AFFECTED (€ M / $ M)	TREATMENT
NOT ADJUSTABLE – RED FLAGS AND DISCLOSURE GAPS · 7			
1	**Provision reversals rise as the result falls (test 1)**	Provisions, 864	Not adjustable: the income line carrying the reversal is not disclosed
2	**Inventories and receivables grow faster than sales (test 2)**	Inventories 6,341, receivables 3,116	Not adjustable: neither allowance is disclosed
3	**Volatile and unreconciled tax charge (test 3)**	Deferred tax assets, 285 (IAS 12)	Not adjustable: no tax reconciliation is published
4	**Impairment spike in 2023 (test 5)**	PP&E, impairment 55	Not adjustable: no test assumptions are disclosed
5	**Composition of other operating expenses not disclosed**	Other operating expenses, 2,170	Not adjustable: the line has no note
6	**No related party disclosure in four years (test 8)**	Whole group	The information is not in the published statements
7	**Customer credit risk carried off balance sheet**	Financial guarantees, not quantified	Not adjustable: no amount is disclosed
REVERSED IN THE WORKBOOK · 5			
8	**Caterpillar inventories on LIFO**	Inventories, equity, operating profit	FIFO excess disclosed – exact correction: $4,305 m
9	**Caterpillar goodwill impairment 2022 inside operating costs**	Operating profit 2022	Moved to other income as a non-recurring item
10	**Liebherr change in inventories and own work shown as income**	Revenue, cost of materials	Both mapped into the cost line (standardised format)
11	**Caterpillar finance receivables inside trade receivables**	Trade receivables	Mapped to other current assets and non-operating investments
12	**Liebherr trade payables not shown on the balance sheet**	Trade payables	Taken from note 7, published subtotal preserved

Source: Report §4.4, Table H2 (short list, 22 findings); Excel workbook, sheets B3, C1, C2, E1 and E2

◂ INDEX

Short list II: what the disclosure does not let us reverse

Findings 13–22 of 22 (Table H2)

#	FINDING	ACCOUNT AFFECTED (€ M / $ M)	TREATMENT
NOT REVERSIBLE – DISCLOSURE TOO THIN · 10			
13	**Gross presentation of rental machine disposals**	Revenue, cost of materials	The amount is not disclosed
14	**The interest element of the finance result**	Finance income 1,043, expenses 1,015	No split is published
15	**No segment note under IFRS 8**	Whole income statement	Segment data only in the management report
16	**The effect of hyperinflation accounting**	Turkish and Ghanaian entities	Not quantified in any year
17	**Actuarial assumptions behind the pension obligation**	Post-employment benefits, 791	No discount rate and no sensitivity disclosed
18	**Inventory and receivable allowances**	Inventories 6,341, receivables 3,116	Neither balance nor charge is disclosed
19	**The two groups do not depreciate at the same rate**	PP&E, 4,304	Measured (Table H3), not carried into the recast – asset mixes differ
20	**Cat Financial inside every Caterpillar figure**	Caterpillar balance sheet	The consolidated basis was set by the assignment
21	**Caterpillar discloses no cost of materials or personnel expense**	Cost structure comparison	Not estimated – operating costs shown as one line
22	**Potential refund of US tariffs after the balance sheet date**	Not recognised; note 12	Non-adjusting event (IAS 10), no reliable estimate

Source: Report §4.4, Table H2 (short list, 22 findings); Excel workbook, sheets B3, C1, C2, E1 and E2

◂ INDEX

Workbook: every number traces back to the report

From the published statements to the ratios – and how we know it adds up

A1–A4

Published statements

typed exactly as printed

B1–B3

Recast by nature

formulas only · 21 mapping decisions

C1–C2

Adjustments

LIFO → FIFO · goodwill 2022 · note data

D1–D11

Analysis

ratios · DuPont · sensitivities · benchmark

E1–E2

Judgements and checks

44 judgements · every subtotal reconciled

Key mapping decisions

- Change in inventories and own work capitalised mapped into the cost of materials
- Caterpillar's finance receivables taken out of trade receivables
- Liebherr's trade payables taken from note 7
- Caterpillar's operating costs kept as one line – no materials or personnel split published
- Liebherr in €, Caterpillar in $ – never added or divided across companies

CHECKS · SHEET E2

0

open differences

Every recast subtotal is a live formula against the published figure. Three deliberate differences are labelled: Caterpillar goodwill 2022, equity in affiliates after tax, and the depreciation difference (judgement 12).

Source: Excel workbook, sheets Method, B1–B3, C1, E1 and E2; Report §4.5 and Appendix K

◂ INDEX

Environment: a modest recovery, but costs rise first

DESTEP scan

Economic

European construction equipment market +4.6 % in 2025, +2 to 2.5 % expected for 2026 – while Liebherr's construction and mining revenue fell 5.5 %.

Political

US tariffs cut Liebherr's North American crane revenue in 2025. Caterpillar expected a tariff impact of $2,600 m in 2026.

Social

Construction had 792 fatal accidents at work in the EU in 2023, 24.0 % of the total. Safety becomes a purchase criterion.

Technological

Autonomous and electrified machines. EU Regulation 2023/1230 covers autonomous functions from 20 January 2027.

Ecological

EU emission limits for non-road machinery and the target of at least 55 % net emission cuts by 2030 make low-emission machines a purchase criterion.

Demographic

Shortages of skilled workers and machine operators push costs up. Personnel expenses reached 28.6 % of revenue in 2025.

Source: Report §3.1.1; CECE (2026), Eurostat (2025), European Union (2023), European Commission (n.d.-a, n.d.-b), Liebherr-International AG (2026), Caterpillar Inc. (2026)

◂ INDEX

Competitive forces squeeze the margin

Porter's five forces

New entrants	Moderate, rising	Chinese producers enter with cheaper, simplified machines. Mobile and crawler crane revenue fell 6.8 % in 2025.
Supplier power	Moderate, rising	Cost of materials is 52.7 % of revenue. Electrification shifts power to battery and electronics suppliers.
Buyer power	High, stable	Receivables rose 9.1 % against revenue growth of 1.0 % in 2025. About half of fleet owners may reconsider their brands by 2028.
Substitutes	Low to moderate, rising	Renting and automation change how much equipment is needed. The rental fleet took €305 m, 28.8 % of 2025 investment.
Rivalry	High, rising	The fixed cost of staying in the market rises. The operating margin fell from 4.6 % in 2024 to 2.8 % in 2025.

INVESTOR LENS **Limited industry profit potential – consistent with the fall to 2.8 %**

Source: Report §3.1.2, Table 1; ratings are the judgement of the report

◂ INDEX

Hard-to-copy resources – locked in the asset base

VRIO and SWOT

VRIO: what is hard to copy

- **Sustained advantage:** in-house key technologies (40+ production sites in 16 countries), application engineering (R&D €708 m, 4.8 % of revenue), family ownership with permanent capital (equity ratio 53.5 %)
- **Parity only:** wholly-owned sales and service network, decentralised divisional structure
- **Temporary to sustained:** internal capital market

Strengths

Vertical integration · equity ratio 53.5 % · 13 segments

Weaknesses

Margin 2.8 % vs. 17.2 % at Caterpillar · operating cash flow €1,171 m → €530 m · no segment disclosure

Opportunities

Low-emission and automated machines · European market +4.6 % in 2025

Threats

Low-price competitors from China · US tariffs · transition costs for new drives and autonomy

Source: Report §3.2 and §3.4, Tables E1 and D1

◂ INDEX

Red-flag screen: reasons to ask, not proof

Nine tests over four years

Four red flags

- **Provision reversals:** €143 m in 2025, 34.6 % of the operating result if credited within it
- **Receivables vs. revenue:** receivables +9.1 % against revenue +1.0 % in 2025
- **Profit vs. tax:** effective tax rate 99.2 %, 45.1 %, 58.2 % and 37.8 % (2022–2025), no reconciliation
- **Impairment:** €55 m in 2023, six times the level of the other years

Clean or not testable

- **Clean:** no off-balance-sheet arrangements disclosed · same auditor (Ernst & Young), no qualified opinion · no change of policy or estimate
- **Not testable:** related party transactions (no note in four years) · large year-end adjustments (no interim figures)

4
flags raised

3
clean

2
not testable

Source: Report §4.4, Table H1; financial statements 2022–2025

◂ INDEX

Where management has room for judgement

Rated by the judgement the standard allows and the materiality of the amount

Highest flexibility

- Inventories: €6,341 m, 36.3 % of total assets, allowance not disclosed
- Provisions: €864 m, of which €530 m warranty
- Impairment test
- Deferred tax assets: €285 m

Also material

- Own work capitalised: €317 m, 77 % of the operating result
- PP&E: €4,304 m, machinery written off over 5 to 21 years
- Revenue timing: advance payments €896 m, contract liabilities €289 m

Lowest flexibility

- Borrowing costs
- Lessee accounting
- Translation of foreign currency
- The standard prescribes the mechanics – little choice for management

Source: Report §4.2–4.3, Table G2 (30 treatments, nine of them key)

◂ INDEX

22 findings: what we could adjust and what we could not

Short list of the accounting analysis

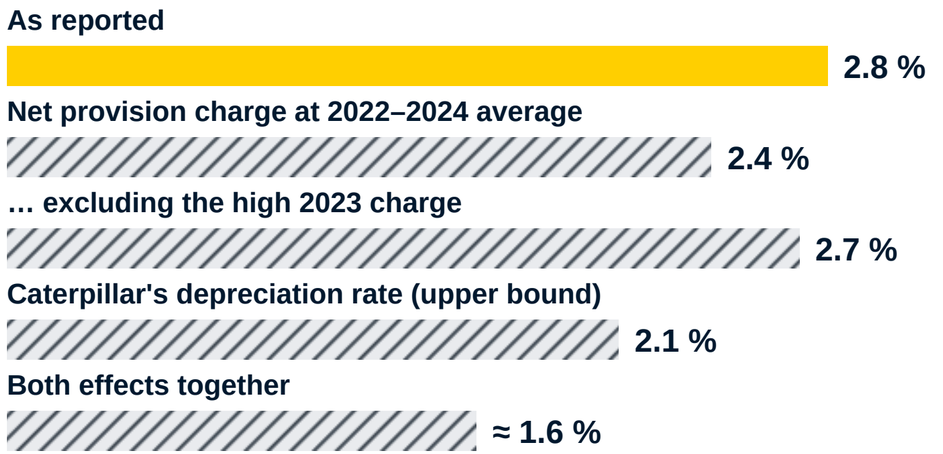
7 **not adjustable**
Red flags and disclosure gaps, e.g. provision reversals of €143 m, missing related party note

5 **reversed in the workbook**
e.g. Caterpillar LIFO → FIFO: equity +$4,305 m, operating profit +$441 m; goodwill impairment 2022

10 **not reversible**
Disclosure too thin, e.g. undefined normal capacity, own work capitalised €317 m

ILLUSTRATIVE SENSITIVITY · NOT BOOKED

Liebherr operating margin 2025



The quantified effects point to a weaker result – the weakness is, if anything, understated.

Source: Report chapter 4, Tables H2, H3 and J13; Excel workbook, sheets C1, D9 and D11

◂ INDEX

How far the key ratios can be trusted

After the accounting analysis the ratios fall into three groups

Reliable

- Operating margin trend
- Inventory and payable days – capital tied up, not asset quality
- Current ratio 2.31, quick ratio 1.16
- OCF / EBITDA 53.0 % and free cash flow −€176 m

Usable with bias

- Margin level 2.8 % – scenarios 1.6 % to 2.8 %
- Days sales outstanding 73.8
- Equity ratio 53.5 % (50.4 % at Caterpillar's depreciation rate)
- Fixed asset turnover 3.49

Not usable

- Net margin 1.8 % and return on equity 3.0 % – tax rate 37.8 % to 99.2 %
- Interest cover – not computable, indicator only
- Comparisons with Caterpillar built on equity

Source: Report §6.3, Table J13; 2025 figures; Excel workbook, sheets D3 and D9

◂ INDEX

Growth and cost structure 2022–2025

Operating margin per year, with revenue and operating profit

2022	2023	2024	2025
3.6 %	**4.4 %**	**4.6 %**	**2.8 %**
Revenue €12,589 m	Revenue €14,042 m	Revenue €14,622 m	Revenue €14,772 m
Operating profit €454 m	Operating profit €619 m	Operating profit €674 m	Operating profit €413 m

Where growth stalled

- Revenue growth 11.5 % (2023) → 4.1 % (2024) → 1.0 % (2025)
- Construction and mining −5.5 % to €9,345 m
- Other segments +14.7 % to €5,427 m, components +35.5 %

Where the margin went

- Personnel expense +5.3 % to €4,226 m vs. revenue +1.0 %
- Personnel share of revenue 26.4 % (2023) → 28.6 % (2025)
- Other operating items cost another 1.25 pp
- Each margin point = €148 m operating profit

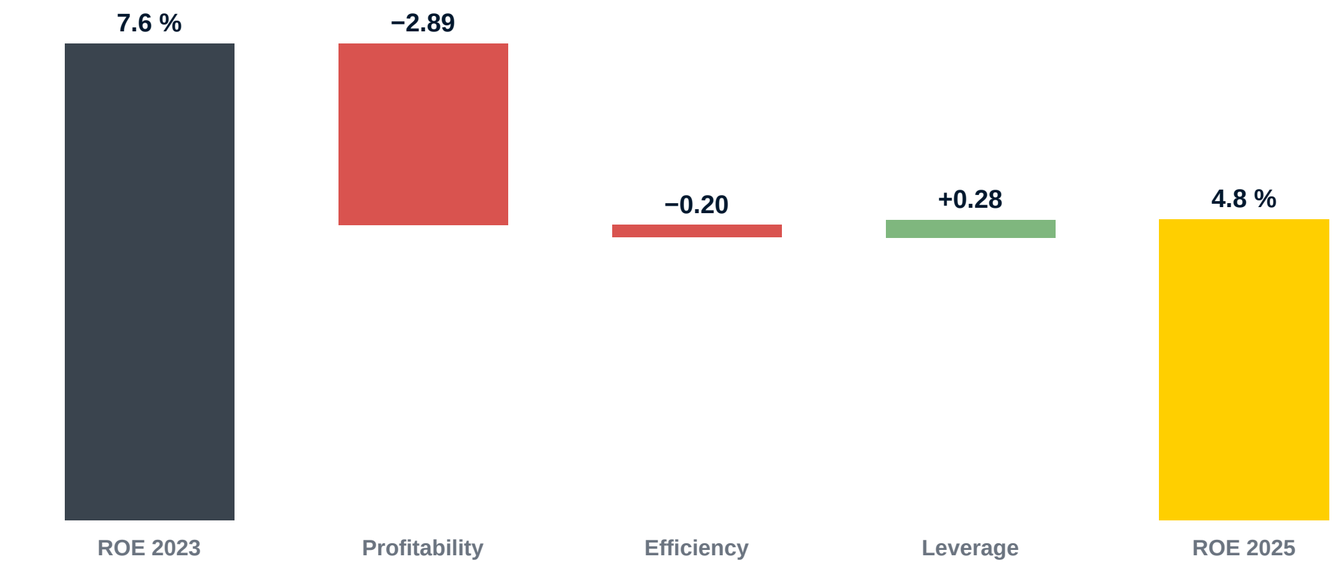
Source: Report §5.2; Excel workbook, sheets D1 and D6

◂ INDEX

DuPont: the margin, not leverage, drives returns down

Pre-tax return on equity 2023 → 2025

Contributions to the change in pre-tax ROE, percentage points



pre-tax return on equity · contributions in pp

What it means

- **Profitability −2.89 pp:** lower pre-tax margin
- **Efficiency −0.20 pp:** slightly lower asset turnover
- **Leverage +0.28 pp:** more debt even helped a little
- After tax, ROE was 3.0 % vs. 16.4 % in the US industry – but the tax charge is volatile, so the trend is read before tax

Source: Report §5.3, Figure 2a, Table J4; Excel workbook, sheets D4 and D6 (logarithmic split)

◂ INDEX

Strong solvency, far weaker returns

Liquidity, solvency and benchmark 2025

Balance sheet strength

- Equity ratio 53.5 %; bank and lease liabilities 23.7 % of capital (industry 39.3 %)
- Cash and current financial assets €3,147 m cover current debt of €1,215 m 2.6×
- Current ratio 2.31 vs. 1.55 at Caterpillar
- Net financial position €523 m (2022) → net debt €85 m (2023) → €249 m (2025)
- Debt / EBITDA 2.14 → 2.90; operating profit / interest paid (indicator) 11.07 (2022) → 3.90 (2025)

2025	LIEBHERR	US INDUSTRY	CATERPILLAR
Operating margin	**2.8 %**	15.9 %	17.2 %
EBITDA margin	**6.8 %**	19.6 %	19.8 %
Return on equity	**3.0 %**	16.4 %	38.1 %*
Inventories, % of revenue	**42.9 %**	16.6 %	33.2 %
Debt / EBITDA	**2.90**	2.30	3.52

* Not comparable – Caterpillar's equity is reduced by treasury shares. Caterpillar on the FIFO basis.

Source: Report §5.1 and §5.5, Table J14; Damodaran (2026b, 2026c, 2026d, 2026e), US machinery industry, January 2026; Excel workbook, sheets D3, D7 and D10

◂ INDEX

Liebherr invests ahead of its cash

Cash flow and capital allocation

190.2 %
Capex in % of operating cash flow, 2022–2025

–€1,151 m
Free cash flow in total, 2022–2025

28.4 %
of EBITDA became operating cash flow, 2022–2025 (53.0 % in 2025)

€530 m
Operating cash flow 2025, down from €1,171 m (2024)

Capex above operating cash flow

- Capex and R&D 9.6 % of revenue vs. 7.4 % at Caterpillar – at a margin of 2.8 % vs. 17.2 %
- Shortfall after distributions €388 m (2023–2025)
- Bank and lease liabilities +€562 m (2022–2025)

Caterpillar returns cash

- Free cash flow $7,453 m in 2025, 11.0 % of revenue
- 73.1 % of operating cash flow returned to shareholders (2022–2025)
- Liebherr distributed €56 m in 2025 – little current income for an investor

Source: Report §5.6, Figure 4; Excel workbook, sheets D5, D7 and D10

◂ INDEX

Why 3.9 % is only a trigger, not a target

Condition 1 of the reassessment

3.9 %
Average operating margin 2022–2025 (3.86 %)

+€157 m
Operating profit at 2025 revenue (€14,772 m)

3.5 %
ROCE before tax 2025 vs. 7.70 % after-tax benchmark

6–8 %
Operating margin needed to earn the benchmark

What the trigger means

- A return to 3.9 % is the first realistic sign of recovery – Liebherr has earned it on average
- It is a test, not a target return
- Each margin point is worth €148 m

Why it is not enough

- Even in 2024, at a 4.6 % margin, ROCE was only 5.9 % before tax – below the 7.70 % benchmark
- The benchmark is the US machinery industry, not Liebherr's own cost of capital
- A margin recovery alone does not prove an adequate return

Source: Report §5.3 and §5.7, chapter 8; Excel workbook, sheets D3 and D10; Damodaran (2026a)

◂ INDEX

What the scenarios are worth

Operating profit at 2025 revenue (€14,772 m) – illustrations, not forecasts

Margin scenarios

2025 actual · 2.80 % margin



€413 m

Four-year average · 3.86 % margin



€570 m **+€157 m**

Best year 2024 · 4.61 % margin



€681 m **+€268 m**

US industry average · 15.86 % margin

€2,343 m **+€1,930 m**

The industry figure shows the size of the gap and is not a target.

Cash scenarios (2025)

- Free cash flow −€176 m: the net financial position of €249 m covers it 1.4 years
- After distributions (−€232 m): 1.1 years
- Capex at the level of depreciation (€587 m): +€119 m, free cash flow still −€57 m
- Cash conversion cycle 10 days shorter: €405 m released once

Source: Report Table J15; Excel workbook, sheet D10. Margins Liebherr has earned or the industry average – nothing is assumed

◂ INDEX

Risks, limits and when to drop the case

What could still go wrong – and the alternative for a cautious investor

Demand

Construction and mining revenue −5.5 % in 2025; no significant upswing expected before 2027

Costs

US tariffs and the cost of autonomous and electrified machines press on the margin

Information

The largest cost increase of 2025 cannot be explained from the summary statements

Valuation

Not assessed in the report – price and contract rights decide the case

Alternative for a risk-averse investor

Equity is unsuitable: low payout, shares not traded. A loan may fit better, as equity of 53.5 % limits the default risk – but pricing, maturity, collateral and covenants are not disclosed.

Drop trigger

If the margin stays near 2.8 % in 2026 **and** the net financial position turns negative, the investor should drop the case rather than wait for 2027. A recovery should not rest on large provision releases.

Source: Report §5.5, §5.7, §6.1 and chapter 8

◂ INDEX

Why not invest now?

Likely questions (1/2)

Q Why not invest now?

Neither financial condition was met in 2025: the margin of 2.8 % is below its four-year average of 3.9 %, and free cash flow was −€176 m. The answer is **not yet**, rather than no.

Q Why is 3.9 % the first test?

It is the average Liebherr earned from 2022 to 2025 – a realistic first sign of recovery. At 2025 revenue it adds €157 m of operating profit. A trigger, not a target return.

Q Is the LIFO to FIFO adjustment fair?

IAS 2 does not permit LIFO, so FIFO puts both groups on one basis. Caterpillar's LIFO reserve of $4,305 m lifts its margin only from 16.50 % to 17.15 % – the gap stays large.

Source: Report chapters 4 to 8; Excel workbook, sheets C1, D3 and D10

◂ INDEX

What the balance sheet cannot fix

Likely questions (2/2)

Q What do the red flags mean?

They are reasons to ask, not proof of manipulation. Four flags are raised, three tests are clean and two cannot be run on the published statements – that is why information rights come first.

Q Why is ROCE so low?

ROCE is 3.5 % before tax against a 7.70 % after-tax benchmark: the margin is only 2.8 %, and inventories of 42.9 % of revenue and a cash cycle 77 days longer than Caterpillar's tie up capital.

Q Why is a strong balance sheet not a return?

An equity ratio of 53.5 % and a net financial position of €249 m limit the risk, but do not earn the cost of capital. They would cover a shortfall like 2025 (−€232 m) for about 1.1 years.

Source: Report chapters 4 to 8; Excel workbook, sheets C1, D3 and D10

◂ INDEX

Questions on method

Likely questions (3/3)

Q Why pre-tax ROE?	The effective tax rate was 99.2 %, 45.1 %, 58.2 % and 37.8 % (2022–2025), with no tax reconciliation published – the after-tax trend says little.
Q Why Caterpillar as the peer?	The world's leading maker of construction and mining equipment – the core market of Liebherr's largest sector (63.3 % of revenue). We compare ratios only, never € and $ figures.
Q Why a US industry benchmark?	Damodaran publishes consistent data for 105 listed machinery companies. It is a reference level, not Liebherr's own cost of capital.
Q R&D expensed, yet €317 m capitalised?	Own work capitalised is self-built assets and rental machines (IAS 16), not development costs (IAS 38), which are never capitalised.
Q Why days on a revenue basis?	Liebherr reports no cost of sales, so DIO and DPO use revenue for both groups – comparable, but not the textbook formula.

Source: Report §4.2, §5.3, §5.7 and §6.3; Excel workbook, sheets C2 and D3; Caterpillar Inc. (2026)

◂ INDEX

References

Sources cited in the presentation, including the backup slides (APA 7th edition)

Caterpillar Inc. (2023). *Annual report on Form 10-K for the fiscal year ended December 31, 2022*. U.S. Securities and Exchange Commission.

Caterpillar Inc. (2026). *Annual report 2025* [Including the annual report on Form 10-K for the fiscal year ended December 31, 2025]. U.S. Securities and Exchange Commission.

Committee for European Construction Equipment. (2026, February 26). *European construction equipment sector sees moderate recovery in 2025 with 4.6% growth in sales*.

Damodaran, A. (2026a). *Cost of equity and capital (US)* [Data set]. NYU Stern School of Business.

Damodaran, A. (2026b). *Debt fundamentals by sector (US)* [Data set]. NYU Stern School of Business.

Damodaran, A. (2026c). *Margins by sector (US)* [Data set]. NYU Stern School of Business.

Damodaran, A. (2026d). *Return on equity by sector (US)* [Data set]. NYU Stern School of Business.

Damodaran, A. (2026e). *Working capital ratios by sector (US)* [Data set]. NYU Stern School of Business.

EUROCONSTRUCT. (2026, June 5). *Press release – 101st EUROCONSTRUCT conference* [Press release].

European Central Bank. (2026, September 10). *Monetary policy decisions* [Press release].

European Commission. (n.d.-a). *Emissions from non-road mobile machinery*.

European Commission. (n.d.-b). *European climate law*.

European Union. (2023). Regulation (EU) 2023/1230 of the European Parliament and of the Council of 14 June 2023 on machinery and repealing Directive 2006/42/EC of the European Parliament and of the Council and Council Directive 73/361/EEC. *Official Journal of the European Union*, L 165, 1–102.

Eurostat. (2025). *Accidents at work – statistics by economic activity*.

IFRS Foundation. (n.d.). *IFRS Accounting Standards Navigator*.

Liebherr-International AG. (2023). *Annual report 2022*.

Liebherr-International AG. (2024). *Annual report 2023*.

Liebherr-International AG. (2025). *Annual report 2024*.

Liebherr-International AG. (2026). *Annual report 2025*.

Own work: Beermann, S., Fliegel, A., Narkiewicz, Z., Hoffmann, P., & Gilmore Islam, Z. (2026). *Liebherr Group financial report: Investment analysis 2022–2025* and the Excel workbook *Liebherr–Caterpillar restructured financial statements*. Fontys University of Applied Sciences, Venlo. · Images: Liebherr-International AG. · Full entries with URLs: report, References.